6.3 Tear & Wear

Wear and tear

Due to continuous working of a machine tool and the nature of work performed on it, wear and tear is observed in the sliding parts of machine tools. As a result of this, the accuracy of the products is affected. In remedy, it is necessary to repair the said machine tools. The affected parts are to be replaced if necessary.

6.4 Backlash

Some slackness is observed in holding of parts, meshing gears or bolt and nut assembly. This slackness is known as backlash. If the amount of backlash is more in the case of mating gears, they should be replaced. The wear and tear on the gears should be observed once in a week time.

6.5 Lubrication

Rotating and sliding parts, which make contact with other parts, are subjected to wear due to friction. Viscous oil called lubricant is applied to these parts to avoid direct contact between them. The process of reducing friction is called lubrication.

Applying oil or grease to the axle shaft of the bullock cart and the cycle wheel, gearbox of automobiles, motor shafts of pumps, fans and sewing machine are some examples of lubrication.

Friction occurs as the shafts are rotating and the sliding parts moving on each other. It generates heat and the parts get damaged.

In order to keep the machine tools accurate and durable, it is necessary to apply lubricants between mating parts. It will reduce friction and wear is minimized.

Lubrication is the nerve centre of machine tool. As it is blood circulation for the human body, lubrication is for the machine tools.

6.6 The machine parts which need to be lubricated

1. Mechanisms of hydraulic systems
2. Guideways and sliding parts
3. Rotating shafts
4. Gear box
5. Feed Box
6. Speed changing mechanisms
7. Bearings

6.7 Purpose of of lubrication

1. Smooth functioning of sliding movable parts
2. To reduce friction and consequent wear
3. To remove burrs and dust
4. To reduce the heat generated due to friction
5. To prevent rust formation on precise parts
6. To provide cushioning effect to the load shocks
7. For hydraulic circuits to transmit power

6.8 Types of lubricants

The materials used for reducing wear between moving and sliding parts are called lubricants. Though there are many types of lubricants available, oil and grease are mostly used.

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